MILITARY SPECIFICATION
CONVERTER GROUP, RADAR SCAN AN/ASA-69

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1.0 SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of performing the functions described herein. The Radar Interface Unit shall accept outputs from the AN/APS-115 Radar, IFF unit, and Data Processing System and convert these outputs to a form usable by the Scan Converter. The Scan Converter shall store this video information for a designated time period to be read-out and displayed on the Multipurpose Displays in one of two separate interlaced rasters, Spiral or TV.

*1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Interface Unit, Radar MX-7974/ ASA-69	3.5.1
Converter, Radar Scan CV-2425/ASA-69	3.5.2
Control, Converter Group, Radar Scan C-7557/ASA-69	3.5.3

*1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2.0 APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids from a part of this specification to the extent specified herein.

FSC5895

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SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-K-25049	Knobs, Control Equipment, Aircraft
MIL-C-26482	Connector, Electric, Circular Miniature, Quick Disconnect
MIL-C-81511	Connector, Electric, Circular, High Density, Quick Disconnect, Environment Resisting, General Specification for

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirement for
WR-101, Part I	Electromagnetic Control Requirements for Advanced ASW Avionics Systems

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STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability Prediction
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements ofr Reliability Program (for Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MS17322	Meter, Time Totalizing, 115 Volt, 400 Cycle

PUBLICATIONS

Military

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment

2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3.0 REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

*3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5).

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

*3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be chosen and approved in accordance with AR-5.

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3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

*3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400, for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 120 pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 40,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement and replacement interval shall be specified by the contractor.

*3.3.2.3 Reliability in Mean Time Between Failure(MTBF)- The specified mean (operating) time between failures for the Radar Scan Converter Group shall be 700 hours, when tested and accepted in accordance with paragraph 4.4.3. This value of MTBF is θ_o as defined in MIL-STD-781.

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3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793:

Unit	Type of Meter
Converter, Radar Scan	MS17322
Interface Unit, Radar	MS17322

3.3.3 Cabling and Connections

*3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. Input power connectors shall conform to MIL-C-26482. Connectors used for digital transmissions shall conform to MIL-C-81511.

*3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781 except that control panels shall be gray per FED-STD-595 (Color No. 36231) with white lights. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

*3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

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*3.3.6 Interference Control - The generation of radio interference by the equipment and vulnerability of the equipment to radio interference shall meet the requirements of WR-101, Part I.

*3.3.7 Provisions for Maintainability - Built-in test equipment, construction and packaging, provisions for test points, and other maintainability parameters shall be in accordance with the intent of AR-10.

*3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 VAC, 3 phase, 400 Hz

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of Specification MIL-E-5400 or ±2g, whichever is less.

*3.3.10.2 Temperature - The equipment shall perform satisfactorily when operated within the temperature range of -25°C to +55°C. Degraded performance will be permitted within the temperature range of -54°C to -25°C, and +55°C to +71°C.

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3.3.11 Warm-Up Time - The time required for equipment warm-up shall be kept to a minimum and shall not exceed 3 minutes under standard conditions, and 5 minutes at -25°C. Turn-on capability shall be maintained down to a temperature of -54°C with no damage to the equipment.

3.3.12 Input Electrical Power

*3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as defined herein. The equipment shall operate within the bounds of limits 2 and 3 of Figure 3 of MIL-STD-704, except that the upper operating limit shall be 130 volts and the lower operating limit shall be 100 volts. Degraded operation may occur when the input voltage exceeds the above limits but remains within limits 1 and 4 of Figure 3 of MIL-STD-704. However, no damage shall be suffered by the equipment and normal operation shall automatically resume upon return of the input voltage to levels within the operating limits as defined above. The power required shall not exceed the following specified amounts:

AC Power - 115 V, Category "B", 600 V.A.3Ø

*3.3.12.2 Lighting Power - Input power for lighting shall require not more than 0.5 amp at 28 volts.

*3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth

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acceptable variations from the performance under the standard conditions will be specified.

*3.4.1 Operation - The Radar Scan Converter Group shall convert, offset, and store the outputs of the AN/APS-115 Radar and IFF unit for presentation on the Multipurpose Display, in response to digital commands from the Data Processing System.

*3.4.2 Registration - When presented on a Multipurpose Display, the converted video output and the raw video output of the Radar Scan Converter Group shall register within 1% of the usable diameter.

*3.4.3 Resolution - The Radar Scan Converter Group shall be capable of storing and converting a minimum of 200 discernible range rings.

3.5 Detailed Requirements

*3.5.1 Radar Interface Unit MX-7974/ASA-69 - The Radar Interface Unit (RIU) shall meet the following requirements:

*3.5.1.1 Function - The RIU shall provide the interface necessary to perform the following functions:

a. The Radar Interface Unit shall accept video, trigger, and azimuth information from the AN/APS-115 and IFF video from the AN/APX-76, and convert these signals to the necessary horizontal and vertical sweeps, video, and unblank signals required for the writing of the radar presentation on the storage tube of Radar Scan Converter CV-2425/ASA-69 and for the display of Raw Radar Video on the MPD.

b. The Radar Interface Unit shall accept data from the Digital Output Multiplexer and the Master Timing Logic, parts of Data Processing System AN/AYA-8, provide the ground stabilization X and Y offset voltages to the radar sweep origin at the write side of the Scan Converter Unit, and execute the commands associated with radar storage and display presentation.

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*3.5.1.2 Form Factor - The Radar Interface Unit shall be enclosed in a single case which has the following maximum dimensions excluding connectors, rubber bumpers, controls, and handles: 18 inches wide, 18-5/8 inches high, and 10-1/4 inches deep.

3.5.1.3 Weight - The weight of the Radar Interface Unit shall not exceed 40 pounds.

3.5.1.4 Contents - The Radar Interface Unit shall consist of the circuit assemblies which comprise the Radar Data Interface and the Ground Stabilization and Command Decoding interface.

*3.5.1.4.1 Radar Data Interface - The Radar Data Interface shall contain the circuit assemblies required to perform the following functions.

*3.5.1.4.1.1 Sweep Generation - Upon receipt of the radar trigger, fixed and resolved sawtooth sweep voltages shall be generated. These sweeps shall be of the proper slope and duration as determined by the selected range scale. These circuits shall include an automatic gate shortening capability which shall provide a fixed "off" time (as required for the recovery of the Scan Converter Unit circuits) and prevent a count down of the PRF for the longer gate periods and the higher PRF. The radar trigger interface circuits shall have a 91 ohm input impedance and be capable of accepting signals between 5 and 10 volts.

*3.5.1.4.1.1.1 Resolved Sweeps - The bearing angle of the Horizontal and Vertical resolved PPI sweeps shall be within ±1 degree of the angle of the 3 wire synchro, 11.8 volt maximum, antenna bearing input, shall be capable of following a 12 RPM input with a velocity error of less than 1/4 degree, shall be capable of sector scanning at rates corresponding to 12 RPM and shall have no greater than 20 percent overshoot for a given input step function. The Horizontal and Vertical sweep

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outputs shall have a fixed baseline, a sweep linearity of less than ±0.5% of maximum range and circularity of less than ±1%.

*3.5.1.4.1.2 Output Line Drivers - Line Drivers shall provide the necessary amplification for the horizontal and vertical sweep signals to drive both the Scan Converter Unit Write Deflection Amplifiers and the MPD Deflection Amplifiers, the latter via the multiplexing circuits.

*3.5.1.4.1.3 Video Processing - These circuits shall accept and mix video signals from both the radar and IFF units and calibrated range marks from an internal generator and shall provide gain compensation for the video such that, for a given signal level, the intensity of the output video from the Scan Converter Unit, as seen on the display, is constant irrespective of the radial distance from the center of the radar display, the radar sweep speed and PRF. At any given gain level (range dependent), the linearity of the video amplification shall be ±5 percent. The video amplifier shall have a minimum bandwidth of 5 megacycles and a maximum of one db droop at a 1500 microsecond pulse width. The output of this circuit shall be capable of driving a 91 ohm terminated line.

The Range Mark Generator shall be capable of generating calibrated range marks. The distance between the range marks shall be such that all selectable range scales shall be easily calibrated. The range circles to appear for the selectable ranges are as shown below.

Range (Miles)	Range Circle Radius (Miles)
8	4
16	4
32	8
64	8
128	16

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The accuracy of these range circles shall be ±0.5 percent of the distance out from the center of the circle.

*3.5.1.4.1.4 Unblank Gate Generation - An unblank gate shall be generated of the proper duration as determined by the selected range scale. This signal shall be supplied as outputs to the Scan Converter Unit write unblank circuits and the MPD unblank circuits. This signal shall also be used by the multiplexing circuits in the "A-Scan" mode. This signal shall be inhibited when Write Radar is not selected, and during update of the ground stabilization voltages.

*3.5.1.4.1.5 Multiplexing - In the raw radar processing function, the RIU shall contain multiplexing switches for selecting either PPI or "A-Scan" radar video presentation. The selection shall be accomplished manually in the test mode or through computer word in the ON-LINE mode. The multiplexed outputs to the MPD shall be as follows:

OUTPUT TO MPD	PPI MODE	"A" SCAN MODE
X Deflection	Horizontal Sweep	Sweep Generator Output
Y Deflection	Vertical Sweep	APS-115/IFF Video
Video	APS-115/IFF Video	Write Unblank
Unblank	Write Unblank	Write Unblank

Each of the multiplexing switches of the RIU shall have the following electrical characteristics: 40 db of crosstalk isolation between PPI and "A-scan" multiplexing switches, and switching speed of not more than 12 microseconds to within 0.1 percent of full scale.

The X and Y output drivers shall have characteristics as per 3.5.1.4.1.1 and 3.5.1.4.1.1.1. The video amplifier shall have characteristics as per 3.5.1.4.1.3. The unblank amplifier shall have characteristics as per 3.5.1.4.1.4.

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*3.5.1.4.1.6 Read Enable Generation - In the On-Line Mode, the RIU shall accept a Read Enable Gate from the Master Timing Logic (MTL) MIL-D-81347 and shall generate a positive 4 ± 1 volt gate of the same width as the input and synchronous with it. In the Test Mode, these circuits shall generate an enable gate, synchronous with the 400 cycle power source, at a frequency of 40 Hz.

*3.5.1.4.1.7 Differential Amplifiers - The Radar Trigger, Read Enable, Radar Video, and IFF Video inputs shall be terminated at the RIU with differential amplifiers. These amplifiers shall provide grounding isolation between the RIU and the aforementioned transmitting sources, and must be capable of tolerating a voltage difference of at least ±4 volts DC between the triax inner connector shield and the airframe. The input impedance of these amplifiers shall be 90 to 95 ohms and 100 pf maximum.

*3.5.1.4.2 Ground Stabilization and Command Decoding Interface - The Ground Stabilization and Command Decoding Interface shall contain the circuit assemblies required to perform the following functions.

*3.5.1.4.2.1 Input Register and Decoding - The RIU shall interface directly with an output channel of the Digital Output Multiplexer (DOM) as per 3.5.1.1 (b). The RIU shall contain input amplifiers as per Appendix I, temporary storage for both ground stabilization and command data, and decoding to provide proper gating of data to the correct circuit assembly. The Digital Output Multiplexer shall transmit data words to the RIU as per Figure 1, with the timing constraints and voltage levels as per Figure 2. Data shall always be present at the Input Amplifiers of the RIU; however, this data shall be sampled only when a "Channel Output Acknowledge" signal is present at the RIU. The "Mode code" portion of the data word shall determine the content of the data word as follows:

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Mode Code	Data
00	Command Code
01	X Offset Code
11	Y Offset Code

The RIU shall inhibit the write unblank signals to the Scan Converter Unit and MPD during that interval of time in which the X and Y offset voltages are being updated.

*3.5.1.4.2.2 X Offset - The RIU shall contain the necessary storage for the X ground stabilization coordinate, and the D/A converter necessary to deflect the radar sweep up to one radius in both the positive and negative X directions. The offset coordinate shall be defined by a nine place binary number, bits 3 to 11 of Ground Stabilization word of Figure 1. The nine place binary numbers are represented in Figure 3 by three place octal numbers, the positive number scale extends from zero to a maximum of 377 octal. Negative binary numbers are represented by corresponding ones complement binary numbers. The octal representation of the ones complement binary numbers extends from octal 777 to octal 400, corresponding to minus zero and minus 377 respectively. The output shall be capable of driving the X off-centering input of the Scan Converter Unit X Write Deflection Amplifier.

*3.5.1.4.2.3 Y Offset - The RIU shall contain the necessary storage for the Y ground stabilization coordinate, and the D/A converter necessary to deflect the radar sweep up to one radius in both the positive and negative Y directions. The number system and D/A requirements are identical to those described in 3.5.1.4.2.2. Normally, the X and Y Offset words will be transmitted within 100 microseconds, but may be transmitted as fast as that shown in Figure 2. During that interval of ground stabilization updating, the Scan Converter Unit and MPD Write Unblank signals shall be disabled.

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*3.5.1.4.2.4 Control Function Decoder - The RIU shall decode bits 3 through 6 of the Command Word, Figure 1, and execute the programmed instruction. The Various "Command Codes" and functions to be performed are shown below.

COMMAND CODE (Octal)	FUNCTION
01	Radar high voltage ON
02	Radar high voltage OFF
03	Erase Scan Converter
04	PPI
05	"A" Scan
06	Store single radar scan
07	8 mile range
10	16 mile range
11	32 mile range
12	64 mile range
13	128 mile range
14	Low PRF (400 cps)
15	High PRF (1600 cps)
16	Store Normal

For each of the command codes, 01, 02, 14, and 15, a separate signal output shall be generated. The characteristics of each output shall be a 6-usec, positive-going pulse. For the command code 03, a 6-usec, negative-going pulse output shall be generated.

When the RIU receives a code of 01, it shall transmit a 6-microsecond pulse to the Radar Power Supply Programmer Unit to turn on the Radar high voltage. When the RIU receives a code of 02, it shall transmit a 6-microsecond pulse to the Radar Power Supply Programmer Unit to turn off the Radar high voltage.

When the RIU receives a code of 03, it shall transmit a 6-microsecond pulse to the Scan Converter Unit Erase Circuit to initiate the erase function.

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When the RIU receives a code of 04, it shall enable the PPI outputs from the RIU multiplexer to be transmitted to the MPD. When the RIU receives a code of 05 it shall enable the "A-scan" outputs from the RIU Multiplexer to be transmitted to the MPD. Either the PPI or "A-scan" outputs will always be at the outputs of the RIU Multiplexer, except that the Unblank output will be inhibited during ground stabilization update and when Write Radar is not selected.

When the RIU receives a code of 06, it shall allow for the storage of one complete radar scan on the Scan Converter Unit. The RIU shall generate a High Voltage On pulse at the beginning of the single scan interval, and a radar High Voltage Off pulse at the end of the interval. The unblank signals shall be disabled at the end of this time period. Single scan shall be possible in both High and Low PRF modes.

Codes 06 and 16 shall enable writing onto the Scan Converter storage tube. For code 16, writing shall be disabled when the RIU receives a code of 02. For a code of 06, writing shall be disabled at the end of the single scan period as described previously.

If the store single scan mode is in progress, and before the single scan has elapsed, the RIU receives a code of 16, the store single scan mode shall be immediately terminated, and the normal store mode shall be immediately initiated.

The codes 07, 10, 11, 12 and 13 signify the range at which the radar presentation is to be stored. The RIU shall decode these codes for proper scaling of the radar presentation. An automatic erase radar shall be performed after receipt of a new range command.

The code 14 signifies that the PRF shall be 400 Hz. The code of 15 signifies a PRF of 1600 Hz. Upon receipt of code 14, the RIU shall transmit a 6-microsecond pulse to the Radar Power Supply Programmer Unit to select the

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low PRF radar mode. Upon receipt of code 15, the RIU shall transmit a 6-microsecond pulse to the Radar Power Supply Programmer Unit to select the high PRF radar mode.

The normal sequence of commands to store the radar presentation on the Scan Converter storage tube is as follows:

a. PRF mode (14 or 15)
b. Range (07, 10, 11, 12, or 13)
c. Radar high voltage ON (01)
d. Store Normal (16)

3.5.1.4.3 Power Supply - The RIU shall contain the power supplies necessary for operation of the analog and digital circuits contained within the RIU.

*3.5.1.4.4 Automatic Preset - At the initial application of primary power, the RIU shall immediately attain the following operational state:

(1) Y offset set to zero
(2) X offset set to zero
(3) 16-mile range
(4) Multiplex set to PPI

The following output pulses shall be generated:

(1) Low PRF
(2) High Voltage OFF
(3) Erase Scan Converter

*3.5.1.4.5 Self-Test Mode - When the self-test mode is manually selected, all inputs from the AN/APS-115 Radar, IFF, MTL, and Digital Output Multiplexer shall be ignored as specified in paragraph 3.5.3.4 and control shall be enabled by the RIU test panel and the Radar Scan Converter Set Control.

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3.5.1.5 Radar Interface Unit Input/Output Signals

3.5.1.5.1 Radar Interface Unit Input Signals - The Radar Interface Unit shall receive inputs from AN/APS-115 Radar, IFF, and Data Processing Unit.

*3.5.1.5.1.1 Inputs from AN/APS-115 - The inputs to the Radar Interface Unit from the AN/APS-115 are:

a. Antenna Bearing - Antenna Bearing inputs will be from a 3-wire synchro, 400 cps, 11.8 volts rms maximum, at a 6 or 12 rpm rate.

b. Radar Trigger - The radar trigger shall be a positive going pulse of 5 to 10 volts amplitude when terminated in 91 ohms. The pulse shall be either 2 ± 0.2 microseconds at a line-locked repetition of 380 to 420 Hz or 0.5 ± 0.1 microsecond at a line-locked repetition of 1520 to 1680 Hz. This signal will be transmitted via triaxial cable.

c. Radar Video - The radar video signal shall be positive-going from ground to plus 2.5 volts when terminated in 91 ohms. This signal will be transmitted via triaxial cable.

*3.5.1.5.1.2 Input from IFF - The input from IFF shall be video, a positive-going pulse of 7.5 ± 2.5 volts peak amplitude when terminated in a 91 ohm impedance. This signal will be transmitted via triaxial cable.

3.5.1.5.1.3 Inputs from Data Processing Unit - Inputs from the Data Processing Unit shall be from two sources; Master Timing Logic and Digital Output Multiplexer.

*3.5.1.5.1.3.1 Inputs from the Master Timing Logic (MTL) - The input signal from the MTL will be the Scan Converter Read Enable. This signal will be a positive-going signal from 0 volts to 4 ± 1 volts, 17.5 milliseconds in width, at a 40 Hertz repetition rate.

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*3.5.1.5.1.3.2 Inputs from the Digital Output Multiplexer - Inputs from the Digital Output Multiplexer will be 11 data lines and one control line. These signals will be transmitted via twisted pair, and each shall be terminated with an Input Amplifier as per Appendix I. The signals shall have the voltage levels and timing constraints of Figure 2.

3.5.1.5.2 Radar Interface Unit Output Signals - Output signals from the Radar Interface Unit shall be to the Scan Converter Unit, Radar Power Supply Programmer Unit, and the MPD.

*3.5.1.5.2.1 Outputs to Scan Converter Unit - Outputs to the Scan Converter Unit are as follows and each shall be transmitted via triaxial cable.

*3.5.1.5.2.1.1 Scan Converter Read Enable - The Read Enable signal shall conform to the following characteristics:

a. On-Line Mode:

Amplitude	4 ± 1 volts
Polarity	Positive
Baseline	0 ± 0.5 volt

Gate Length and frequency shall be identical to the input defined in 3.5.1.5.1.3.1

b. Test Mode

Amplitude	4 ± 1 volts
Polarity	Positive
Baseline	0 ± 0.5 volt
Frequency	Iterative every 25 milliseconds in synchronization with the 400 Hz power source.
Gate Length	Variable between 12.5 and 22.5 milliseconds.

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*3.5.1.5.2.1.2 Horizontal Sweep - The Resolved Horizontal Sweep signal shall be adjustable from zero to between ±6 to ±8 volts when terminated in 91 ohms. The output voltage shall not drift more than 40 millivolts.

3.5.1.5.2.1.3 Vertical Sweep - The Vertical Sweep signal shall have the characteristics as per 3.5.1.5.2.1.2.

*3.5.1.5.2.1.4 Write Unblank - The Write Unblank signal shall be positive-going from 0 ± 0.5 to plus 5 ± 1 volts and shall have a rise time of less than 0.5 microsecond when terminated in 91 ohms.

*3.5.1.5.2.1.5 Write Video - The write video signals to the storage tube shall be positive, with an amplitude which varies proportional to the input amplitude over the input range of zero to plus 2.5 volts. The video shall be compensated to meet the requirements of 3.5.1.4.1.3.

*3.5.1.5.2.1.6 X and Y Offset Signals - Each of the X and Y Offset Signals shall be levels between zero and a maximum which is adjustable between ±6 and ±8 volts where the maximum represents ±1 radius. The amplitude of these signals shall be proportional to the binary numbers as defined in 3.5.1.4.2.2 and 3.5.1.4.2.3 and shall be stable to within 40 millivolts when terminated in 91 ohms.

*3.5.1.5.2.1.7 Erase - The Erase signal shall be a 6 ± 3 microsecond negative-going pulse, where 0 ± 0.5 volt represents a logical "one" (erase) and plus 4 ± 1 volts represents a logical "zero", when terminated with 91 ohms.

3.5.1.5.2.2 Outputs to Radar Power Supply Programmer Unit - Outputs to the Radar Power Supply Programmer Unit (PSPU) are as follows:

*3.5.1.5.2.2.1 Radar High Voltage ON - The Radar High Voltage ON signal shall be a positive-going 6 ± 3 microsecond pulse, where 4 ± 1 volts represents a logical "one" (high

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voltage ON) and 0 + 0.5, -0.0 volt represents a logical "zero". This signal shall be transmitted to the PSPU by twisted pair and shall be terminated with an input amplifier, as per Appendix I.

3.5.1.5.2.2.2 Radar High Voltage OFF - The Radar High Voltage OFF signal shall have characteristics identical to the signal of 3.5.1.5.2.2.1.

3.5.1.5.2.2.3 Low PRF Radar Mode - The Low PRF radar mode signal shall have characteristics identical to the signal of 3.5.1.5.2.2.1.

3.5.1.5.2.2.4 High PRF Radar Mode - The High PRF radar mode signal shall have characteristics identical to the signal of 3.5.1.5.2.2.1.

*3.5.1.5.2.3 Outputs to MPD - The outputs to the MPD shall be transmitted via triaxial cable terminated in 91 ohms and shall consist of the following:

3.5.1.5.2.3.1 X Deflection - In the PPI mode, the X deflection output shall be identical to that of 3.5.1.5.2.1.2. In the "A-scan" mode, the X deflection output shall be a linear ramp from zero to +6 to +8 volts maximum.

3.5.1.5.2.3.2 Y Deflection - In the PPI mode, the Y deflection output shall be identical to that of 3.5.1.5.2.1.3. In the "A-scan" mode, the Y deflection output shall be identical to that of 3.5.1.5.1.1 (c), except that the signal bandwidth shall be 1 MHz.

3.5.1.5.2.3.3 Video - In the PPI mode, the video output shall be identical to that of 3.5.1.5.1.1 (c). In the "A-scan" mode the video output shall be identical to that of 3.5.1.5.2.1.4.

3.5.1.5.2.3.4 Unblank - In both the PPI and "A-scan" modes, the unblank output shall be identical to that of 3.5.1.5.2.1.4.

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3.5.1.6 Cooling - The Radar Interface Unit shall be designed to operate in a temperature environment of +55°C maximum. If air circulating fans are required for operation at this temperature, they may be installed in the Radar Interface Unit. The method of cooling must be approved by the procuring activity prior to release for fabrication.

3.5.1.7 Mounting - The Radar Interface Unit shall have provisions for being hard mounted.

*3.5.1.8 Controls - A Test Panel on the RIU shall contain the controls as described herein.

(1) Power - ON Normal/OFF
(2) Raw Radar A-Scan/PPI
(3) Command Selection
    a. 400/1600 PRF
    b. HV ON/OFF
    c. Erase
    d. Offset
    e. Storage
    f. Raw Radar
    g. Registration
(4) PRF 1600/400
(5) HV ON/OFF
(6) Offset X/Y
(7) Offset
    a. -R
    b. -R/2
    c. -0
    d. +R/2
    e. +R
(8) Store Normal/Single
(9) Enter

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*3.5.1.8.1 Power-On Normal/Off Switch - The Power On switch shall provide power to the Radar Interface Unit and switches mounted on the Radar Scan Converter Set Control.

*3.5.1.8.2 Command Selection Switch - This rotary switch shall enable the various command modes so that they may be entered into the RIU in the TEST mode. Only the switch associated with the selected command shall be energized to provide input data to the RIU; all other test switches except the Enter switch shall be disabled. No data is entered into the RIU until the Enter switch is depressed.

*3.5.1.8.2.1 400/1600 PRF Position - When the Command Selection switch is in this position, the 400/1600 PRF switch shall be enabled.

*3.5.1.8.2.2 HV ON/OFF Position - When the Command Selection switch is in this position, the HV ON/OFF switch shall be enabled.

*3.5.1.8.2.3 Erase Position - When the Command Selection switch is in this position, the Erase Code (03) may be entered into the RIU by depression of the Enter switch.

*3.5.1.8.2.4 Offset Position - When the Command Selection switch is in this position, the X Offset/Y Offset Switch and the Offset rotary switch shall be enabled.

*3.5.1.8.2.5 Storage Mode Position - When the Command Selection switch is in this position, the Storage switch shall be enabled.

*3.5.1.8.2.6 Registration Position - When the Command Selection switch is in this position, an internal registration pattern generator shall be enabled when the Enter switch is depressed. The pattern shall be simultaneously supplied to both SCU and MPD outputs. The pattern shall consist of an X inscribed in a square whose sides are approximately 3/4 of a tube diameter. The deflection waveforms and unblanking gates

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making up the pattern are generated during the inactive portion of the read enable gate period (7.5 ms). Triggered from the negative-going edge of the read enable gate, the timing period consists of a 2.0 ms delay followed by 8 strokes of approximately 0.5 ms per stroke.

The pattern is switched by multiplexing gates onto the horizontal and vertical deflection and unblank output lines of the RIU (and disabling the normal output signals) which feed the Scan Converter Unit and Multipurpose Display.

*3.5.1.8.2.7 Raw Radar Position - When the Command Selection switch is in this position, the Raw Radar A-Scan/PPI switch shall be enabled. Whether in the Test or On-Line mode, inputs from the AN/APS-115, as per 3.5.1.5.1.1, shall be processed and transmitted to the MPD, as per 3.5.1.4.1.5.

*3.5.1.8.3 400 PRF/1600 PRF Switch - This switch shall be a toggle-type where either the 400 PRF or 1600 PRF radar mode shall be active. When the 400 PRF is selected, the code of 14 shall be entered into the RIU upon depression of the Enter switch. When the 1600 PRF is selected, the code of 15 shall be entered into the RIU upon depression of the Enter switch. In both cases, the proper 6-microsecond pulse to the Radar Power Supply Programmer Unit shall occur upon receipt of either input code. This switch shall also control the RPM of the resolved sweeps.

*3.5.1.8.4 HV On/HV Off Switch - This switch shall be a toggle-type where either the HV ON or HV OFF mode shall be active. When the HV ON is selected, the code of 01 shall be entered into the RIU upon depression of the Enter switch. When the HV OFF is selected, the code of 02 shall be entered into the RIU upon depression of the Enter switch. In both cases, the proper 6-microsecond pulse to the Radar Power

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Supply Programmer Unit shall occur upon receipt of either input code.

*3.5.1.8.5 X Offset/Y Offset Switch - This switch shall be a toggle-type where either the X Offset or Y Offset mode shall be active. When the X Offset is selected, the "mode code" of binary 01 as per Figure 1, shall be present at the RIU. When the Y Offset is selected, the "mode code" of binary 11 as per Figure 1, shall be present at the RIU. These codes, one at a time, shall be entered in conjunction with the selected offset position by depression of the Enter switch.

*3.5.1.8.6 Offset Switch - This rotary switch shall provide the capability of selecting one of the five offset positions along each axis, or a total of 25 different offset positions as per Figure 4. Each offset position shall be entered by first selecting the X Offset mode of the X Offset/Y Offset switch, selecting the desired X Offset position from the Offset switch, and depressing the Enter switch to enter the data into the RIU and then by repeating the same process with the X Offset/Y Offset switch in the Y Offset mode.

*3.5.1.8.7 Storage Switch - The switch shall be a toggle-type where either the NORMAL or SINGLE scan mode shall be active. When this switch is in either of the two positions, the appropriate code shall be present at the inputs of the RIU. The Code shall be entered into the RIU by depression of the Enter switch.

*3.5.1.8.8 Enter Switch - This is a momentary switch, that when depressed, shall enable all data from the test panel present at the inputs to the control function decoder of the RIU to be entered into the RIU.

*3.5.1.8.9 Raw Radar Switch - This switch shall be a toggle-type where either the PPI or A-SCAN mode shall be active. When this switch is in either of the two positions,

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the appropriate code shall be present at the inputs of the RIU. The code shall be entered into the RIU by depreseion of the Enter switch.

*3.5.2 Radar Scan Converter CV-2425/ASA-69 - The Scan Converter Unit (SCU) shall meet the following requirements

3.5.2.1 Function - The SCU shall be capable of storing electronically written information for specified periods of time. It shall provide the video and deflection outputs for presentation of the stored information on Multipurpose Displays. The SCU shall be capable of the simultaneous storage and readout as explained herein.

*3.5.2.2 Form Factor - The Scan Converter Unit shall be enclosed in a case which has the following maximum dimensions including controls and connectors but excluding mounting flanges and latch covers: 15 inches high, 10 inches deep, and 31.5 inches wide. Connector and control locations must be approved by the procuring activity prior to release for fabrication.

3.5.2.3 Weight - The weight of the Scan Converter shall not exceed 75 pounds.

*3.5.2.4 Contents - The Scan Converter Unit shall consist of a Storage Tube assembly and circuit assemblies necessary for writing the RIU processed radar video into the storage tube. The SCU also shall contain the circuits required for reading the stored information and presenting this to the MPD.

*3.5.2.4.1 Write Deflection Amplifiers - Each deflection amplifier shall be capable of accepting RIU

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horizontal and vertical sweep outputs, 3.5.1.5.2.1.2 and 3.5.1.5.2.1.3. The output of these amplifiers shall deflect the write beam of the Scan Converter's storage tube via the write magnetic deflection coil. The response and settling time to within one percent of one diameter shall not exceed 10 microseconds. Inputs shall also be provided for the X and Y Offset voltages as per 3.5.1.5.2.1.6.

*3.5.2.4.2 Write Video Processing - The SCU shall be capable of accepting the compensated video output of the RIU, 3.5.1.5.2.1.5. These signals shall be amplified and further compensated to match the storage tube characteristics prior to being applied to the storage tube. These circuits shall have a bandwidth of 20 megahertz.

*3.5.2.4.3 Write Unblank Circuits - The SCU shall be capable of accepting unblank signals from the RIU (3.5.1.5.2.1.4) The signal applied to the write side of the storage tube shall have a stability of ±100 millivolts for the unblanking portion of the unblank signal. The rise time of the tube unblank signal measured from the 10 percent to the 90 percent points, shall be less than 0.5 microsecond.

*3.5.2.4.4 Storage Tube Erase - The SCU shall accept input signals from the RIU, 3.5.1.5.2.1.7, and perform the necessary electronic operations to erase and prime the storage element of the Scan Converter storage tube. During the erase time, which shall be no greater than two seconds, the write unblank and read video output shall be inhibited. It shall also be possible to manually initiate the erase cycle from the SCU front panel.

*3.5.2.4.5 Storage Tube and Circuits - The storage of the radar information shall be accomplished in a dual gun storage tube. This tube shall be capable of storing either

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a single scan of radar, or successive scans continuously written and integrated on the storage screen. Appropriate circuitry shall be provided to allow for storage of the radar information for a variable time as controlled by the setting of the "Storage Time" control on the Radar Scan Converter Set Control, 3.5.3. The storage time shall be adjustable from 10 seconds to 300 seconds over the temperature range of -25°C to +25°C and shall derate linearly over the temperature range of +25°C to +55°C to a maximum storage time of 120 seconds at +55°C. Storage time shall be defined as that time in which the information output decays to 50 percent of its initial value, which shall be 90% of saturation, after the source of information has been removed, under the condition of continuous readout.

*3.5.2.4.6 Read Deflection Amplifiers - The Read Deflection Amplifiers shall control the deflection of the read beam during the Read mode of operation. The time to deflect the beam one diameter ±1 percent, shall not exceed 10 microseconds. The phase shift of the deflection current, when reading at a 16KC rate, shall be continuously adjustable over a range of ±6 degrees, minimum.

*3.5.2.4.7 Read Output Signals - Two sets of read output signals (horizontal raster, vertical raster, read unblank, and output video) shall be provided via triaxial connectors.

*3.5.2.4.8 Read Scan Generation - The reading raster of the SCU shall be gated by the Read Enable from the RIU and shall be a spiral scan. The frequency shall be 16 KHz ± 0.5% over the temperature range. The sine and cosine components of the raster shall reverse polarity in succeeding enable time periods yielding a spiral raster with a 2 to 1 interlace. The enable time period shall be as per 3.5.1.5.2.1.1. An

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alternate set of sweep generators shall generate horizontal and vertical components of an interlaced TV raster. The horizontal scan shall be 40 microseconds, with a maximum of 10 microseconds retrace time. This set of sweep generators shall be interchangeable with the spiral scan generators. Only one type of sweep generator shall be used per Scan Converter at any one time. The horizontal and vertical components of the selected raster shall be supplied as output signals for the MPD deflection.

*3.5.2.4.9 Read Video Output - The SCU video outputs (one for each MPD) shall be present during the read enable period only. These outputs shall be inhibited during the erase cycle. The signal-to-noise ratio of these outputs shall be a minimum of 15 to 1. The bandwidth of the read video shall be sufficient to maintain a resolution of 800 TV lines per diameter.

*3.5.2.4.10 Read Unblanking - The storage tube reading gun shall be unblanked in the following manner, depending on the type readout (Spiral or TV) employed.

*3.5.2.4.10.1 Spiral Readout - In the Spiral Readout mode the read unblank waveform shall be gated on by the read enable. This signal shall be shaped so that the read beam current is approximately proportional to the read beam linear velocity.

*3.5.2.4.10.2 TV Readout - In the TV Readout mode, the read unblank shall be gated on by the read enable. The read beam intensity shall be reduced during horizontal retrace to insure a uniform brightness scan converter output.

*3.5.2.4.11 Focus - The SCU shall contain the static and dynamic focus circuits required to maintain the specified resolution over the useable area of the storage tube surface.

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*3.5.2.4.12 High Voltage Timing Circuits - These circuits shall provide timing delays as necessary for the High Voltage Power Supply (HVPS). The application of power to the input of the HVPS shall be delayed to allow for the filaments of the Scan Converter storage tube to reach the proper operating temperature. During turn-on and turn-off transients, read and write unblank signals shall be inhibited.

*3.5.2.4.13 Power Supplies - The Scan Converter Unit shall contain the low voltage power supplies which shall provide the necessary voltages for the operation of the solid state circuitry, and high voltage power supplies which shall provide the biasing and accelerating voltages for the Scan Converter storage tube.

*3.5.2.4.14 Differential Amplifiers - The Horizontal and Vertical Sweeps, X and Y Offsets, Write Video, Write Unblank, Erase, and Read Enable inputs shall be terminated at the Scan Converter Unit with differential amplifiers. These amplifiers shall provide the necessary grounding isolation between the Scan Converter Unit and the transmitting sources. Electrical characteristics of this circuit shall be as per 3.5.1.4.1.7.

3.5.2.5 Resolution - The resolution of the Scan Converter Unit outputs shall be such that a minimum of 200 discernible concentric centered range rings shall be obtainable.

3.5.2.6 Registration - Registration between the read and write positions shall be less than 1 percent of the usable diameter.

3.5.2.7 Scan Converter Unit Input/Output Signals

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*3.5.2.7.1 Scan Converter Input Signals - The Scan Converter Unit shall receive inputs from the Radar Interface Unit.

*3.5.2.7.1.1 Inputs from Radar Interface Unit - Inputs from the Radar Interface Unit are as follows and shall be transmitted via triaxial cable.

3.5.2.7.1.1.1 Scan Converter Read Enable - The signal shall be as defined in 3.5.1.5.2.1.1.

3.5.2.7.1.1.2 Horizontal Sweep - This signal shall be as defined in 3.5.1.5.2.1.2.

3.5.2.7.1.1.3 Vertical Sweep - This signal shall be as defined in 3.5.1.5.2.1.3.

3.5.2.7.1.1.4 Write Unblank - This signal shall be as defined in 3.5.1.5.2.1.4.

3.5.2.7.1.1.5 Write Video - This signal shall be as defined in 3.5.1.5.2.1.5.

3.5.2.7.1.1.6 X and Y Offset Signals - These signals shall be as defined in 3.5.1.5.2.1.6.

3.5.2.7.1.1.7 Erase - This signal shall be as defined in 3.5.1.5.2.1.7.

*3.5.2.7.2 Scan Converter Output Signals - The Scan Converter shall transmit output signals to each of the two Multipurpose Displays, AN/ASA-70, via triaxial cable.

*3.5.2.7.2.1 Read Video - The Read Video signal shall be positive-going from 0 ± 0.5 volt to 2.5 ± 0.5 volts, and shall have the characteristics as defined in 3.5.2.4.9, and shall be capable of driving a 91 ohm terminated line.

*3.5.2.7.2.2 Horizontal Deflection Output - The Horizontal Deflection output shall be adjustable from zero

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to between ±6 to ±8 volts maximum, and shall have the characteristics as defined in 3.5.2.4.8. The base line for the spiral output shall be zero volts ±40 millivolts. The output of this circuit shall be dc coupled and capable of driving a 91 ohm terminated line.

3.5.2.7.2.3 Vertical Deflection Output - The Vertical Deflection output shall have the characteristics as per 3.5.2.7.2.2.

*3.5.2.7.2.4 MPD Unblanking Output - The MPD Unblanking output shall have the characteristics of the Scan Converter Read Enable, defined in 3.5.2.7.1.1.1, except that in the TV mode the signal shall be gated-off during the horizontal retrace time.

3.5.2.8 Cooling - The Scan Converter Unit shall operate in a temperature environment of +55°C maximum. If air circulating fans are required for operation at this temperature, they may be installed in the Scan Converter Unit. The method of cooling must be approved by the procuring activity prior to release for fabrication.

3.5.2.9 Mounting - The Scan Converter Unit shall have provisions for being hard mounted.

*3.5.2.10 Controls - The SCU shall contain the following controls.

*3.5.2.10.1 Power-ON Normal/OFF - This toggle-type switch shall control the application of AC power to the SCU.

*3.5.2.10.2 Erase - This shall be a momentary type switch, which when activated shall initiate the storage tube erase cycle.

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*3.5.3 Radar Scan Converter Group Control

C-7557/ASA-69 - The Radar Scan Converter Group Control shall meet the following requirements:

*3.5.3.1 Function - The Radar Scan Converter Set Control shall contain controls for varying the storage time of the Scan Converter storage tube as per 3.5.2.4.5, selecting ON-LINE or TEST mode, selecting the entering Radar range in TEST mode, and selecting Range Rings.

*3.5.3.2 Form Factor - The Radar Scan Converter Set Control shall consist of four switches and one control mounted on an illuminated pannel as per MIL-C-6781, except for color which shall be grey per FED-STD-595 (color No. 36231). The electronics associated with these controls shall be protected by a dust cover attached to the rear of this panel. All connectors shall be mounted on the rear of this unit. The control knobs shall conform to MIL-K-25049, and shall be illuminated by white lamps used to light the plastic lighting plate markings, as per MIL-P-7788. The size of the unit shall not exceed 5-3/4 inches wide, 5-1/4 inches high and 6 inches deep overall.

*3.5.3.3 Weight - The weight of the Radar Scan Converter Set Control shall not exceed five pounds.

*3.5.3.4 Controls - The Radar Scan Converter Set Control shall contain operator controls as listed below:

a. Storage Time - This control shall be used to regulate the storage time for information to be stored on the Scan Converter storage tube. The limits of this control shall vary the storage time from 20 to 300 seconds. The control shall provide fineness of adjustment

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in the short store range.

b. Range Switch - This rotary switch shall provide the capability for the selection of one of five radar ranges. The selected range shall be entered into the RIU upon depression of the Enter switch on the Radar Scan Converter Group Control under the condition that the ON-LINE/TEST switch is in the TEST position. The range codes are as per 3.5.1.4.2.4.

c. Range Enter Switch - This is a momentary switch, that when depressed, shall enable the radar range data to be entered into the RIU. The switch legend, RANGE ENTER, normally shall be the lit green in the TEST mode, but when depressed shall light amber. The background shall not be illuminated. This switch shall not be illuminated in the ON-LINE mode.

d. Range Rings Switch - This switch shall be an alternate-action-type. When this function is active the switch legend, RANGE RINGS, shall be lit amber and when inactive the legend shall be lit green. The background shall not be illuminated. When this switch is in the active mode, the range rings as defined in 3.5.1.4.1.3 shall be written onto the storage tube of the Scan Converter Unit in both the ON-LINE and TEST modes. When inactive, no range rings will be stored in either mode. In the ON-LINE mode, the number and separation of these rings shall be determined by the range code received from the computer via the DPS. In the TEST mode, the number and separation of the rings shall be determined by the range code entered from the Radar Scan Converter Group Control.

e. On-Line/Test Switch - This switch shall be an alternate-action-type, where either the ON-LINE or the TEST mode shall be active. The legend for the active mode shall be lit amber, and the legend for the inactive

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mode shall not be illuminated. The background shall not be illuminated. When the ON-LINE mode is selected the Radar Scan Converter Group shall operate in a slave mode to the DOM. When any Test mode except Raw Radar is selected, all external imputs shall be ignored and the Radar Scan Converter Grou½ shall respond to inputs from the test panel of the RIU and the Radar Scan Converter Group Control. When in the Raw Radar Test Mode, inputs from the AN/APS-115 as per 3.5.1.5.1.1, shall be processed and transmitted to the MPD, as per 3.5.1.4.1.5.

*3.5.3.5 Cooling - Cooling of the Radar Scan Converter Group Control shall be accomplished by natural convection.

*3.5.3.6 Mounting - The Radar Scan Converter Group Control shall be mounted on a standard control panel mounting rail assembly with 5-3/8 inch mounting centers, as per MIL-C-6781.

*3.5.3.7 Switch Performance - The performance of the alternate and momentary action switches on the Radar Scan Converter Group Control shall be equivalent to the Master Specialty series 90E switches.

*3.5.3.8 Grounding - All electronics contained in the Radar Scan Converter Group Control shall be isolated from airframe ground.

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4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the procuring activity

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

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*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all requirements of this specification and the contract. Preproduction tests shall include Environmental Tests in accordance with the procedures of Specification MIL-T-5422.

*4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction samples. Prefabrication of production equipment prior to the approval of the preproduction samples shall be at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to the submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

*4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction samples. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the procuring activity, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests.

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No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample and that it meets all applicable requirements including design, construction, workmanship, performance and quality.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors

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not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The Contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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*4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of ten (10) hours without failure. A failure shall be defined as any thing which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustment points as are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test included a test on each equipment which consumes at least 10 hours of operation.

Temperature - Ambient room
Humidity - Ambient room
Vibration - Any selected frequency within the range of 20 to 60 cps (excluding resonana points) and a minimum amplitude of ±3g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten (10) minutes prior to ghe beginning of the 10-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of the vibration should be vertical to the normal mounting plane for five (5) minutes and lateral to the plane for five (5) minutes. Where it is not feasible to vigrate the equipment in two (2) directions, the vertical direction shall be used. Dur9ng the ten (10) hour period of operation following the ten (10) minute vibration period, the equipment shall be cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten (10) minute vibration period need not be repeated when it is certain the failure was not a result

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of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten (10) hour period specified above may be composed of two five (5) hour periods to conform with standard working hours.

*4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0#
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

#NOTE: If by contract action the Initial Production Test (4.3) is deleted, a sample Test shall be conducted on one (1) equipment from the first ten (10) produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action proposed.

*4.4.3 Reliability Assurance Tests - Reliability Assurance tests shall be conducted as required by MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (sampling) Phase shall be conducted.

*4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used except that the lower temperature limit will be -25°C. The Accept-Reject Criteria for Test Plan IV shall be used.

*4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - Samples of the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production

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Reliability Tests". For the Sampling Phase (or All-Equipment Test), Test Level E shall be used except the lower temperature limit shall be -25°C.

*4.4.3.2.1 All Equipment Tests - Each equipment produced shall be operated until 30 consecutive hours of failure free operation have been accummulated. Test Level E of MIL-STD-781, with a lower temperature limit of -25°C, shall be used for this test. A failure shall be defined as any condition which causes an equipment malfunction. Only adjustment of those controls which are accessible to the operator during normal operation will be permitted.

*4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

*4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will

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be determined by the procuring acitivity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.
(2) Determine if the failure is an isolated case or design defect.
(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.
(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which

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have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

*4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
AC Voltage	115 ± 5 volts (at lowest applicable frequency)
DC Voltage	27.5 ± 2.0 volts

*4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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*4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring acitivity.

*4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

*4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the

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procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

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5.0 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6.0 NOTES

6.1 Intended Use - The equipment covered by this specification shall be capable of converting, offsetting, and storing radar video information for presentation on the TACCO and Sensor Operator Multipurpose Displays in a P-3C aircraft.

*6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (see 5.1)

*6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification

(2) This Specification - This specificatio shall have precedence over all applicable subsidiary specifications.

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Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

*6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a

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Mean-Time-To-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.6. Type Designations - The parentheses () when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

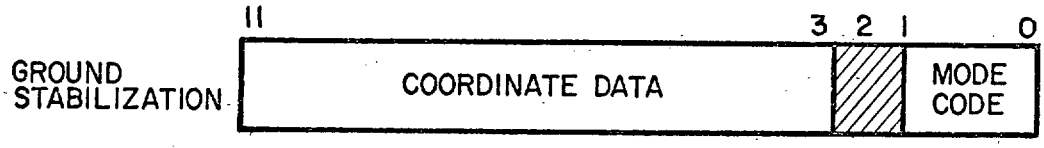
6.7 Revisions - In specification revisions and superseding amendments an asterisk "" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

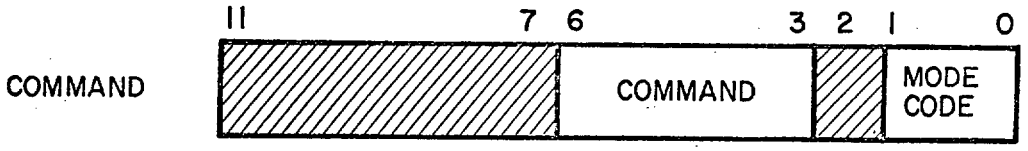
*6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

AN/APS-115	Radar
AN/APX-76	IFF
AN/AYA-8	Data Processing System
AN/ASA-70	Multipurpose Display

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0	0	COMMAND WORD
0	1	X OFFSET WORD
1	1	Y OFFSET WORD

FIGURE 1. DATA WORDS FROM DIGITAL OUTPUT MULTIPLEXER

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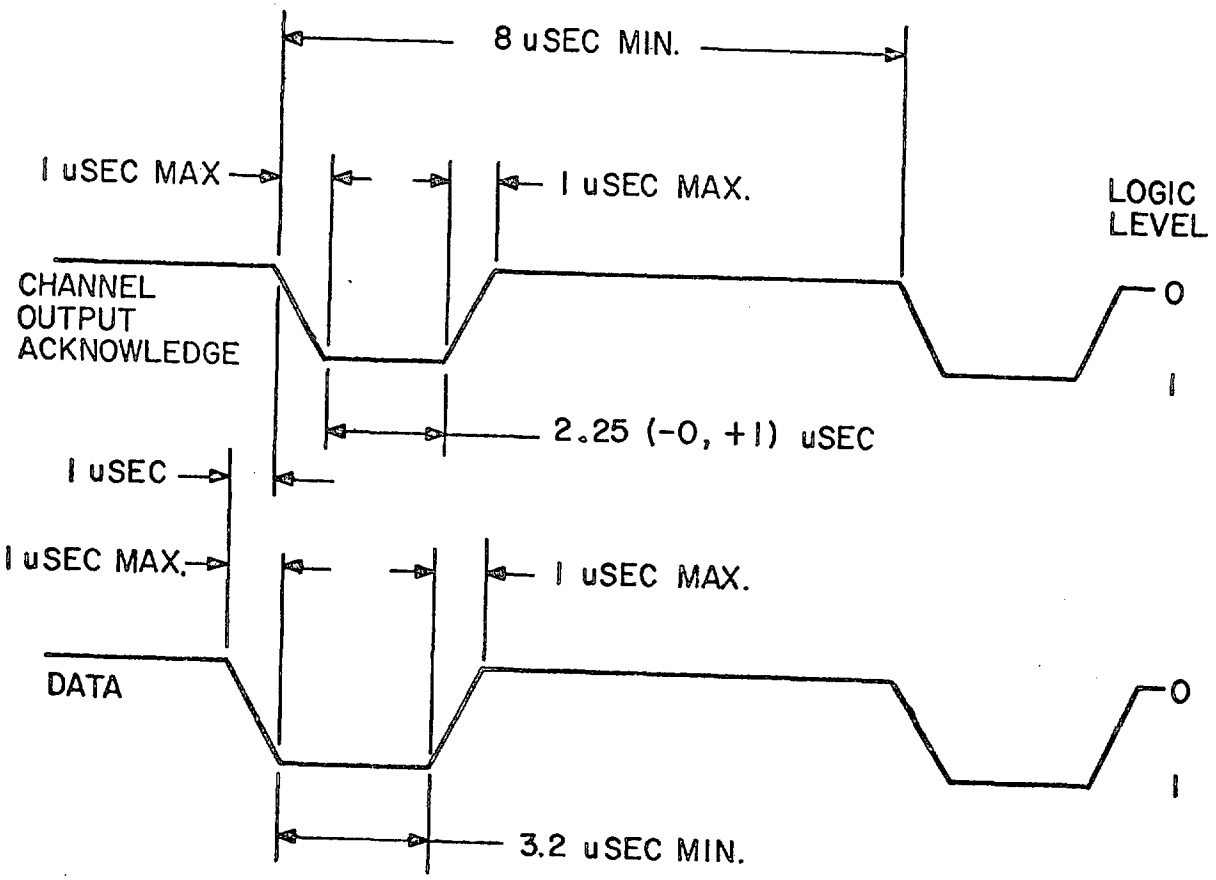


FIGURE 2. TIMING SIGNALS FROM DIGITAL OUTPUT MULTIPLEXER

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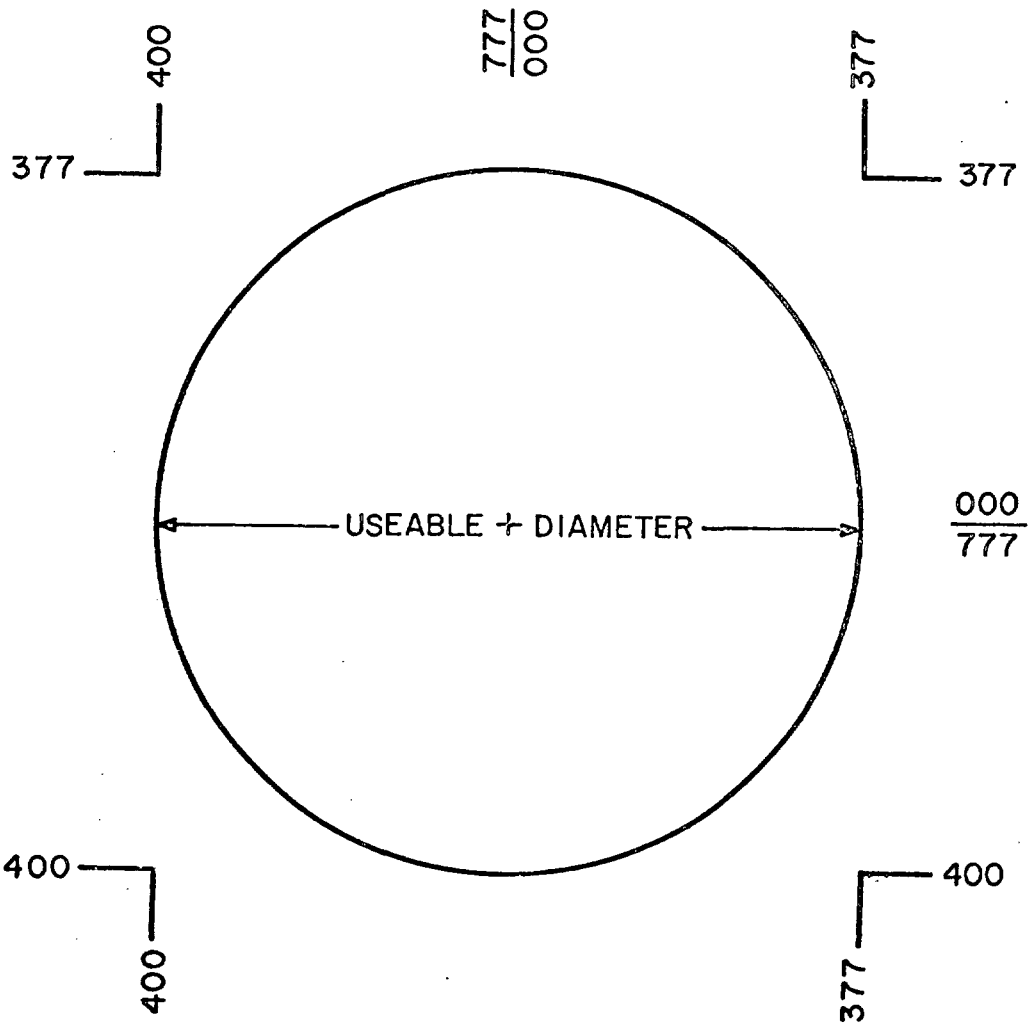


FIGURE 3. X AND Y COORDINATE SYSTEM IN OCTAL CODE

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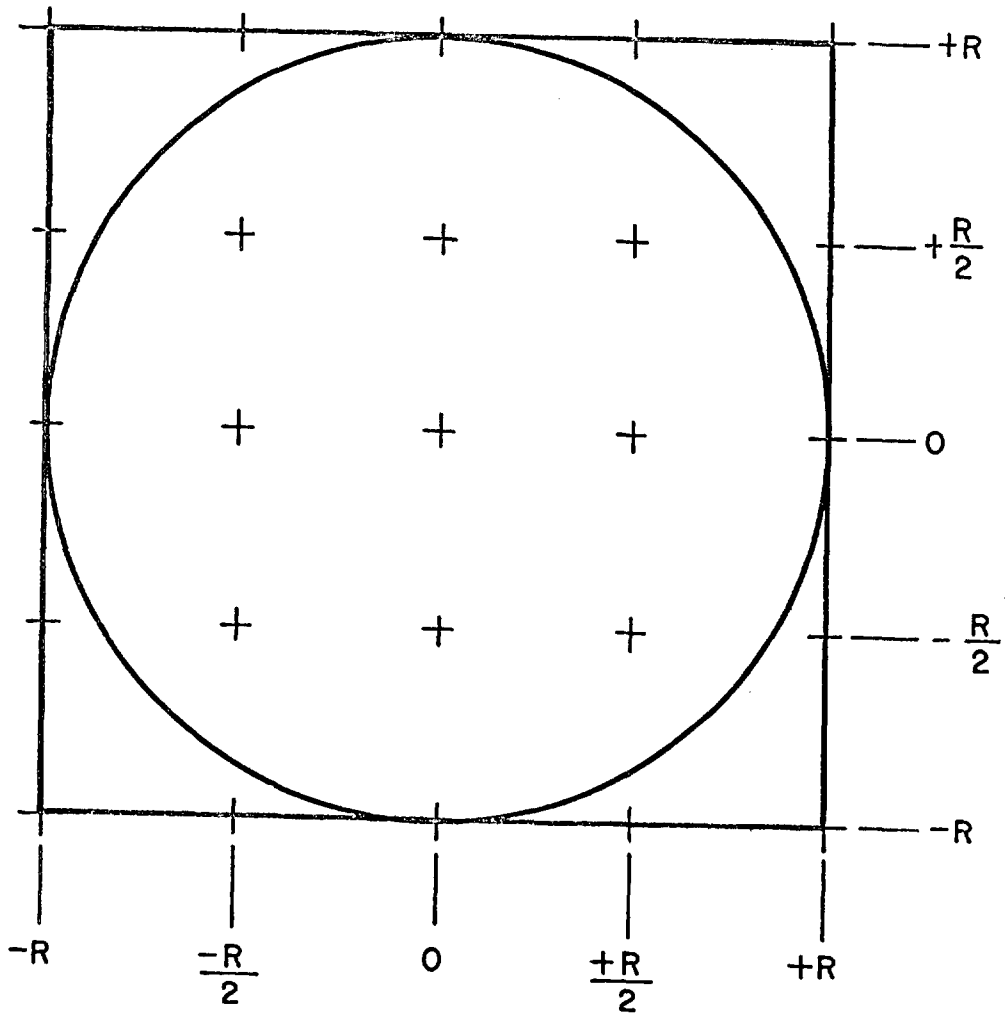


FIGURE 4. TEST MODE OFFSET POSITIONS

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APPENDIX I

Digital Input Amplifier Characteristics - The circuit characteristics of the Digital Input Amplifiers shall be as follows:

(1) Voltage Level Inputs - The logic "one" state will be 0 ± 0.5 volt and the logic "zero" state is +4 ± 1 volts as measured at a Data Line Driver in the Data Processing System.

(2) Trigger Threshold - The threshold level distinguishing the "one" state from the "zero" state is +1.5 ± 0.5 volts at the Input Amplifier terminal.

(3) Noise Rejection - The circuit shall be designed for ±5 volts common mode noise rejection. It shall be capable of handling noise levels up to ±5 volts peak, without any change in output level.

(4) Rise and Fall Times - The time required for a data line driver of the DPS to switch logic levels at the output terminals of the driver is no greater than 1.0 microsecond.

(5) Loading - The maximum steady state current drawn from a line by an input amplifier circuit shall not exceed 1.5 milliamperes.

(6) Input Impedance - The Input Amplifier circuit shall have an AC input impedance of 130 ohms and a DC input impedance of 2000 ohms.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81343C(AS) CONVERTER GROUP, RADAR SCAN AN/ASA-69

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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